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How to Model an IT Management Landscape and its Transformation using the ArchiMate®, IT4IT™, and TOGAF® Standards

A Case Study by:

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July 2019

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How to Model an IT Management Landscape and its Transformation using the ArchiMate®, IT4IT™, and TOGAF® Standards
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Table of Contents

Executive Summary **4**

Introduction **5**

Introducing ArchiSurance 6

Introducing the ArchiMate Model of the IT4IT Reference Architecture 7

Modeling a Transformation of an IT Management Landscape **8**

Consolidation of IT Service Request Portals 10

Baseline State: Six IT Requests Portals 11

Transition State 1: New IT Service Request Portal 14

Transition State 2: PRO-FIT IT Service Request Portal 16

Transition State 3: One Cloud IT Service Request Portal 18

Target State – Step 4: One Cloud-Based IT Service Request Portal 20

Modeling an IT Management Landscape **22**

Conclusion **24**

APPENDIX 1: Mapping the IT4IT Reference Architecture to the ArchiMate Language **25**

IT4IT Level 1 Class Model 25

IT4IT Level 2 Class Model 26

IT4IT Level 3 Class Model 28

Additional Considerations 28

The Solution-Pattern 33

APPENDIX 2: ArchiSurance Health Assessment using the IT4IT Reference Architecture **35**

References **40**

About the Author **41**

About The Open Group **42**



Boundaryless Information Flow ™
achieved through global interoperability
in a secure, reliable, and timely manner

Executive Summary

This Case Study is a fictitious example developed to illustrate the use of the IT4IT™ Reference Architecture together with the ArchiMate® modeling language, both standards of The Open Group, through the following two use-cases:

- Modeling a transformation of an IT Management Landscape
- Modeling an IT Management Landscape

This Case Study is an extension of the existing ArchiSurance Case Study. Even though the concepts presented in this Case Study could be applied to different situations, the example addresses typical concerns corresponding to the insurance and financial sectors.

This work supports The Open Group vision of Boundaryless Information Flow™ by illustrating the combined use of the ArchiMate and IT4IT standards. This document assumes a basic understanding of the ArchiMate and IT4IT standards.

Introduction

IT departments of large enterprises can use the IT4IT™ Reference Architecture, a standard of The Open Group, to understand the correctness and completeness of the set up of the *Business of IT* by comparing its own IT architecture with the Reference Architecture. Through this comparison, the enterprise can answer questions such as:

- Is our current IT architecture fit-for-purpose?
- Can we identify gaps and duplications in the needed functionality and data set to help improve our own IT architecture?

At the same time, the ArchiMate® modeling language, a standard of The Open Group, can be used to model the business, using the language visualization capability to represent the different layers of the Enterprise Architecture (as defined within the TOGAF® framework, a standard of The Open Group).

Where the ArchiMate language can be seen as generic, allowing any organization to be modeled, the IT4IT Reference Architecture can be seen as specific for the IT department within an organization. Where the ArchiMate language allows the architecture team to model all layers of the architecture as defined by the TOGAF standard, supporting the whole Architecture Development Method (ADM), the IT4IT Reference Architecture focuses in detail on the logical functional components and the data set managed.

Although the IT4IT Reference Architecture uses a bespoke modeling approach to visualize the reference architecture, it does not prescribe the meta-model (see Chapter 4 of the IT4IT Reference Architecture, Version 2.1) beyond the elements it needs for the representation, leaving the expansion of the model into the other parts of the Enterprise Architecture to the user of the Reference Architecture.

This opens up the question of how this can be done. How can the IT4IT Reference Architecture be expanded beyond the functional and data architecture? Given the fact that both the ArchiMate and IT4IT standards are provided by The Open Group, it is only natural to explore this expansion by combining the two. Two perspectives often emerge:

- The perspective of looking at modeling an IT landscape
- Given that such a landscape is never static, the perspective of looking at the transformation of the current mode of operations of a landscape into a future mode of operations

This Case Study is a fictitious example developed to illustrate the use of the IT4IT Reference Architecture together with the ArchiMate modeling language through the following two use-cases:

- Modeling a transformation of an IT Management Landscape
- Modeling an IT Management Landscape

To make these use-cases realistic, this document uses a virtual company called ArchiSurance.[1] ArchiSurance is an insurance company that has recently been through a merger of three companies and, as a result, needs to transform its IT operations to run as a single organization and thereby provides the context of the two use-cases mentioned above.

Introducing ArchiSurance

The text in this section is taken from the ArchiSurance Case Study, Version 2 (see References).

ArchiSurance is the result of a merger of three previously independent insurance companies:

- *Home & Away*, specializing in homeowners' insurance and travel insurance
- *PRO-FIT*, specializing in auto insurance
- *Legally Yours*, specializing in legal expense insurance

The company now consists of three divisions with the same names and headquarters as their independent predecessors, as shown in Figure 1.

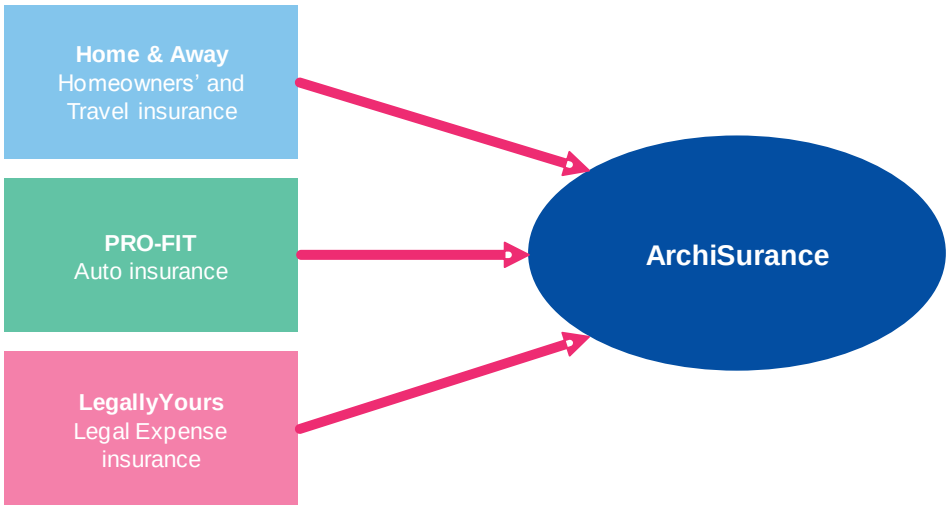


Figure 1: The Merger that Created ArchiSurance

ArchiSurance was formed to take advantage of numerous synergies between the three organizations. While the three pre-merger companies sold different types of insurance, they had similar business models. All three sold direct to consumers and businesses through the web, email, telephone, and postal mail channels. Although based in different cities, each was completely housed in a modern office complex in a major metropolitan area. Each had loyal customer bases and strong reputations for integrity, value, service, and

[1] Archisurance was selected because it is the company used by The Open Group ArchiMate Forum to support the explanation of the usage of the ArchiMate language, reducing the need to introduce a new company.

financial stability. All three companies were privately held by interlocking groups of institutional and individual investors.

The lead investors of the three companies began merger talks after they noticed that lower-cost competitors were entering their markets, that there were new opportunities in high-growth regions, and that each company required significant new IT investments to remain competitive. They realized that only a larger, combined company could simultaneously control its costs, maintain its customer satisfaction, invest in new technology, and take advantage of emerging markets with high growth potential. The merger negotiations and regulatory approvals took 18 months, but two years ago the papers were signed and the merger was complete.

The new company offers all the insurance products of the three pre-merger companies, and intends to frequently adjust its offerings in response to changing market conditions. Like its three predecessors, ArchiSurance sells directly to customers via web, email, telephone, and postal mail channels, and indirectly via intermediaries.

Introducing the ArchiMate Model of the IT4IT Reference Architecture

The Open Group provides a model of the IT4IT Reference Architecture using the ArchiMate modeling language. This document takes this model as the basis for the expansion, answering the two questions.

Appendix 1 provides an overview of the mapping of the Level 1 to 3 class models to ArchiMate objects and the additional considerations that have been used for using the IT4IT and ArchiMate standards to model a transformation scenario.

Modeling a Transformation of an IT Management Landscape

After the merger, ArchiSurance wants to be able to benefit from the merger and, as indicated, invest in IT. To enable the investments to be the most efficient and at the same time be able to support the business in its daily operations, ArchiSurance IT Management has commissioned a health assessment on the current state of the IT after the merger. Appendix 2 provides a high-level overview of the results of this merger, using the IT4IT Reference Architecture as the basis for the assessment. Using these results, ArchiSurance IT Management has prioritized five initiatives based on their expected contribution to post-merger goals. These initiatives are realizing improvements in IT's ability to meet customer expectations, cost reductions, improvement of external service provider compliance, and improvement of operational performance.

The following initiatives are prioritized:

1. **Service Portfolio Consolidation (S2P)** – Identification of the current state of used IT services across the heritage companies mapped to their function, determination of overlap in IT services for required functions, and the definition of the target state and functionality use of IT services. This initiative realizes the IT organization's most important post-merger goals. It will identify the IT services the business is consuming and consolidate them in alignment with ArchiSurance business goals and available resources. The initiative will also aim to reduce the number of IT service providers, enabling ArchiSurance to better manage external service provider compliance, lower the IT cost, and help ArchiSurance in meeting their customers' expectations.

   This initiative will have the highest priority and therefore starts first, since it will provide required insight for other initiatives and will realize the most significant cost reduction benefits next to freeing up IT resources and attention for subsequent efforts.

2. **IT Service Request Portal Consolidation (R2F)** – Through the merger, ArchiSurance currently has six heritage IT Service Request Portals. Maintaining these six heritage IT Service Request Portals is costly and the possibility of providing consolidated IT services to the business will be difficult. Therefore, ArchiSurance desired future state is to have a single IT Service Request Portal providing the single entry point into IT. This initiative will select a new (cloud-based) IT Service Request Portal system, consolidating the six heritage systems, connected to the IT service catalogs and request fulfillment systems.

3. **Configuration Management Service Consolidation (D2C)** – The pre-merger companies each had one or more Configuration Management System (CMS) systems. Given the merger, targets require operational improvements and cost reduction. Being able to understand and support the IT services with multiple CMS systems is costly and creates errors in the operations. The desired future state is therefore to consolidate the CMS systems into one. This one system must be capable of providing a consolidated view of both the cloud and on-premise systems. Given PRO-FIT had already consolidated its cloud and on-premise configuration management, ArchiSurance divisions derived from Home & Away and Legally Yours will migrate to the solution from PRO FIT. For this, five CMS systems will need to consolidate to the one system from PRO FIT.

4. **Service Desk Rationalization (D2C)** – Before the merger, each company had its own helpdesk system. Going forward, maintaining all these systems is costly and operational excellence is difficult to achieve when IT helpdesk employees need to work in multiple systems for single tasks. Preferred is a helpdesk system which can be kept "evergreen" with minimal effort. The desired future state, therefore, is the set-up of a consolidated Software as a Service (SaaS)-based service desk system – migration from the two on-premise heritage helpdesk systems to one SaaS helpdesk system already in use by ArchiSurance divisions coming from PRO-FIT. This will create a single helpdesk capability with a "follow-the-sun"[2] support model with helpdesks in North America, Europe, and Asia. This new global capability will use the consolidated CMS.

5. **Service Management Consolidation (D2C)** – ArchiSurance has three heritage event management and monitoring capabilities, one in each of the regions for Europe, North America, and Asia, reducing the ability for ArchiSurance to achieve true global support and operational efficiency. ArchiSurance will move to a "follow-the-sun" support model using a single event management and monitoring capability to be able to benefit from a lower-cost and improved operation efficiency. Three heritage service management systems will consolidate into one global centralized system with three regional operations centers. The global system will be used by the unified service desk capability and will also be connected to the consolidated CMS so it can process events in context.

Figure 2 shows that each of the prioritized initiatives contributes to the goals set by ArchiSurance IT Management.

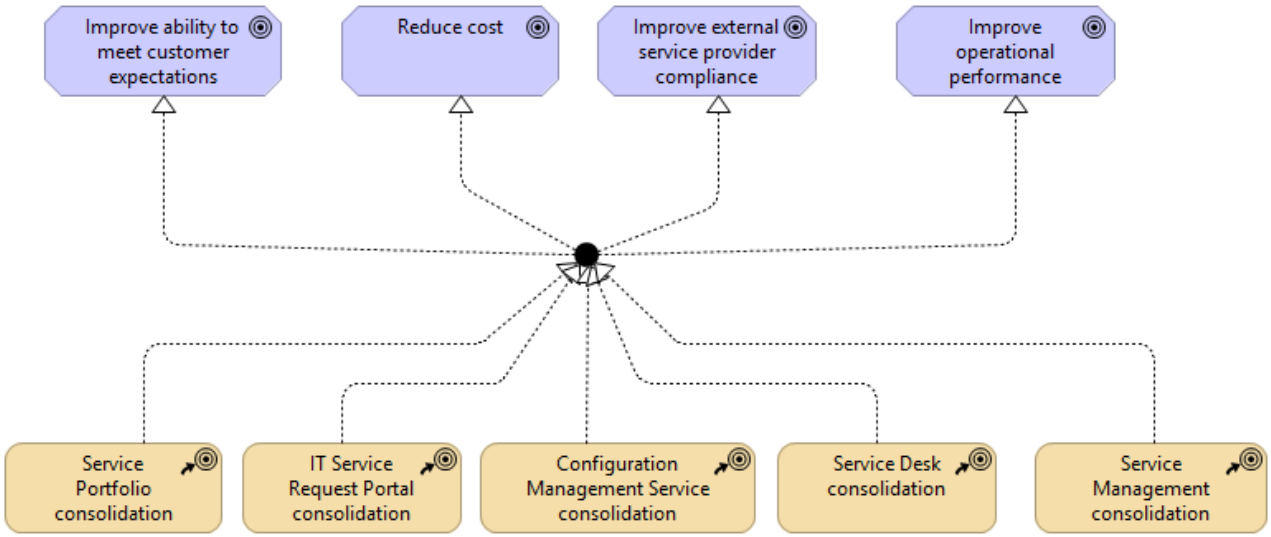


Figure 2: ArchiSurance Motivation and Strategy for IT Service Management Initiatives

[2] "Follow-the-sun" support model: a support model that provides round-the-clock service support using a global workflow model that follows the regional time zone-based working hours for (more local) support.

This document will be limited to the second scenario, which focuses on the consolidation of the six IT Service Request Portals.

Consolidation of IT Service Request Portals

ArchiSurance has three heritage IT Service Request Portals for on-premises services, one from each heritage company, and three IT Service Request Portals provided by cloud service providers as each heritage company also used a separate IT service provider for cloud services. In order to begin realizing the IT consolidation efficiencies, as promised to investors as part of the merger, ArchiSurance IT Management must enable one new enterprise-wide portal for requesting and approving IT services and must decommission its current heritage portals.

ArchiSurance IT Management has additionally decided on the following guiding principles. The new system needs to be able to:

- Support all types of IT requests

  Given IT Technology Management at ArchiSurance in time wants to consolidate its cloud service providers and migrate completely to the cloud, the consolidated IT Service Request Portal must thus support both cloud and on-premises infrastructure and platform requests, as well as other types of requests, such as those arising from software and hardware incidents.

- Be provided using an SaaS model

  Given building and managing IT Service Request Portals should not be the core business of ArchiSurance and ArchiSurance IT, both want to be able to benefit from new developments in the market as quickly as possible.

- Work with multiple internal and external service providers

  Given ArchiSurance does not want to execute a large migration of the service delivery to prevent business interruption, all existing internal and external service providers need to be connected as-is. Furthermore, ArchiSurance will continue to expand the usage of external service providers and wants to prevent vendor lock-in, the system needs to be able to integrate easily with new internal and external fulfillment engines.

Taking all this into consideration, the decision is taken to consolidate all six heritage IT Service Request Portals into one provided by a cloud service broker called ArchiSoft, connected with the existing set of fulfillment engines. This allows ArchiSurance to continue its delivery and operations of its services and at the same time allow addition and removal of internal and external service delivery.

To reduce the risk of the consolidation, ArchiSurance IT Management have decided the consolidation will be done in phases using transition states. They envisage the following transition steps:

- Transition Step 1: engage ArchiSoft to provide a unified portal for requesting cloud services

  This step is a pre-requisite for the transformation. It is aimed to get access to the portal and allow a Proof-of-Concept (PoC) to validate the business case and the technology selected.

- Transition Step 2: migrate the PRO-FIT heritage portal to the new cloud-based unified portal

  By moving the first set of cloud services (of the PRO-FIT cloud service providers) to the new portal, this

step will allow ArchiSurance to start aligning the external service providers and improve the service compliance as well as learn from the migration.

- Transition Step 3: migrate the Home & Away and Legally Yours heritage portals to the new cloud-based portal for cloud services

  Moving all cloud services to one portal will allow ArchiSurance to continue to improve the service compliance of its external service providers.

- Target State – Step 4: migrate the Home & Away and Legally Yours heritage portals to the new cloud-based portal for cloud services

  Use an Application Programming Interface (API) to provide a single IT Service Request Portal using the ArchiSoft Engagement Experience Portal and supporting components.

Figure 3 is an ArchiMate Implementation & Migration view that shows this progression and how it contributes to ArchiSurance post-merger goals.

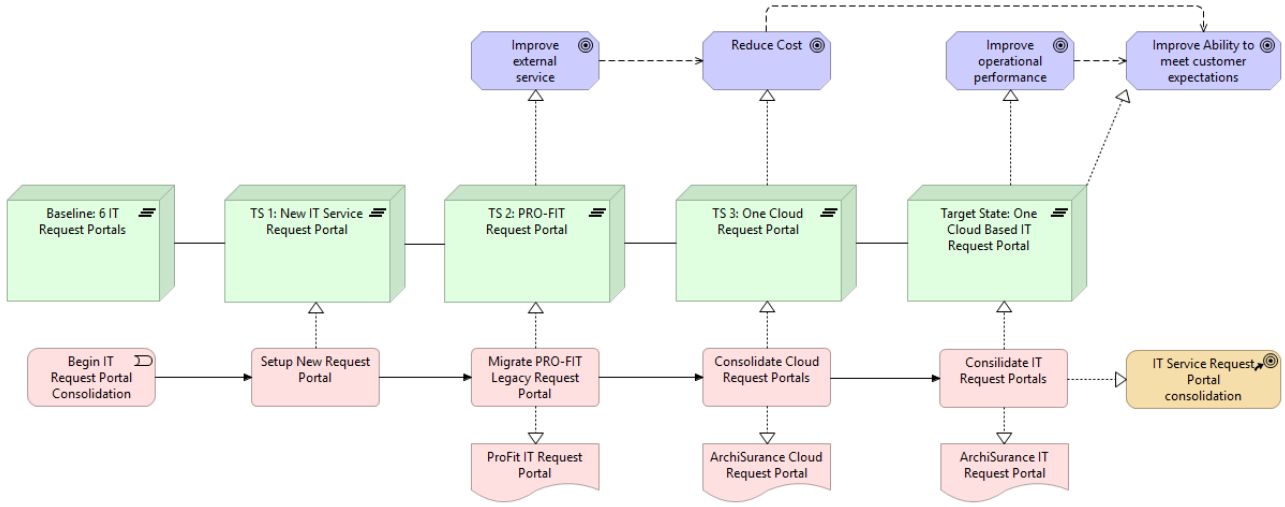


Figure 3: ArchiSurance IT Service Request Portal Consolidation Implementation & Migration View

Figure 3 uses ArchiMate plateaus to represent the baseline, transition, and target states of the ArchiSurance IT Service Request Portal consolidation effort. The effort consists of a series of projects represented as ArchiMate work packages. Each work package realizes a plateau and some of the plateaus achieve a (part of) a goal. ArchiMate implementation events denote the initiation and completion of the overall effort.

Baseline State: Six IT Requests Portals

The assessment using the IT4IT Reference Architecture (see Appendix 2) revealed that after the merger, ArchiSurance has a total of six IT Service Request Portals. Each company uses its own IT Service Request Portals, providing traditional IT services as well as cloud services both separately set up to support requesting of traditional IT services and cloud services from external cloud service providers.

To better understand the current state, ArchiSurance IT Management architects relate the heritage information systems and modules to the IT4IT functional components which are part of the Request to Fulfillment (R2F) value stream, as listed in Table 1:

Table 1: Key IT4IT Functional Components used in this Scenario

Functional Component Name	Functional Component Definition
Offer Consumption	Present service offers to various service consumers. Facilitate consumption/management of and payment for IT services rendered.
Request Rationalization	Rationalize, break down, and route "clean order" requests (ready for fulfillment) to appropriate Fulfillment Execution engines or providers in order to deliver services to consumers.
Fulfillment Execution	Orchestrate the delivery of the various requests amongst (one or more) fulfillment engines in order to deliver the IT service.

In Figure 4, the result of the mapping is provided. As can be seen, in order to support the business, the number of components to maintain is relatively high.

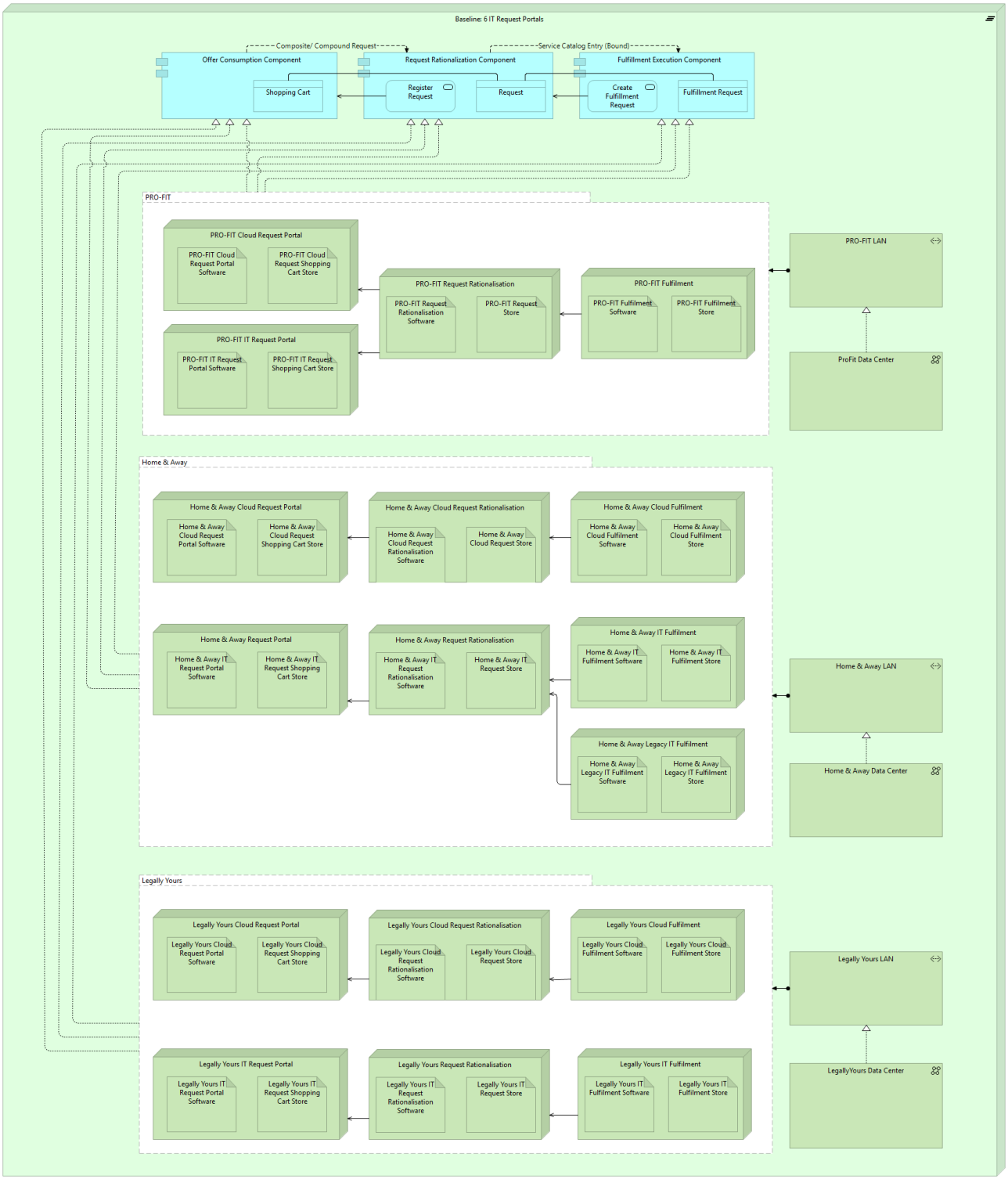


Figure 4: Baseline State: Six IT Service Request Portals

Transition State 1: New IT Service Request Portal

In the first step to reach Transition State 1, the new cloud-based ArchiSoft IT Service Request Portal is implemented. Once this plateau is complete, the new system is available for executing a Proof-of-Concept (PoC) and Proof-of-Value (PoV). Using the system in production is not possible because it is not yet connected to any fulfillment engine.

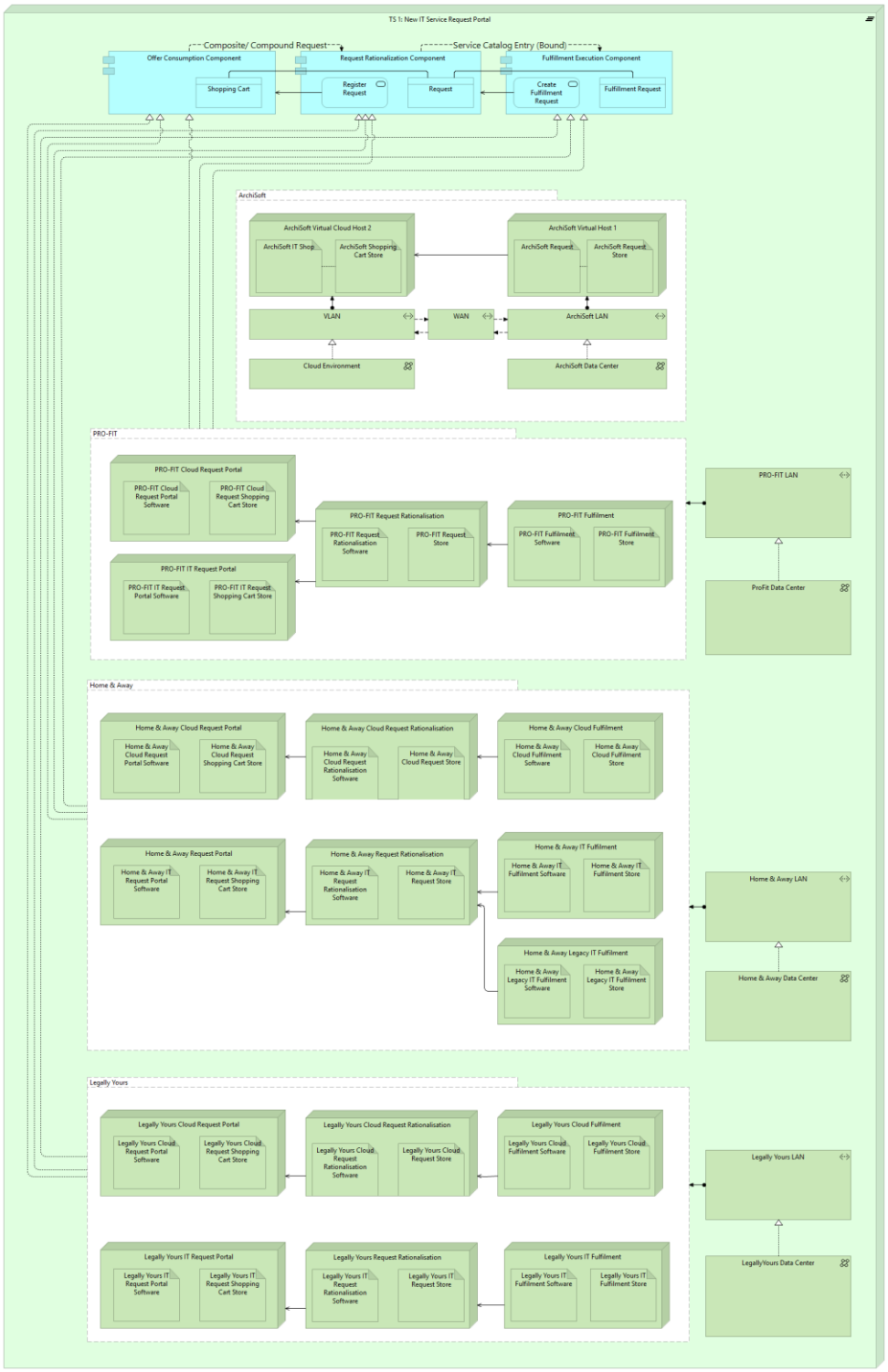


Figure 5: Transition State 1: New IT Service Request Portal

Transition State 2: PRO-FIT IT Service Request Portal

As the new IT Service Request Portal is in place, the next migration step is to replace the complete PRO-FIT cloud and IT Service Request Portals with the new cloud-based ArchiSoft IT Service Request Portal and connect the PRO-FIT implemented Fulfillment Execution component (PRO-FIT Fulfillment Software) with the new implementation of the ArchiSoft Request Rationalization component (ArchiSoft Request). From this moment onwards, the IT Shop of the heritage PRO-FIT users requesting IT services (internal and cloud) is the new ArchiSoft IT Shop.

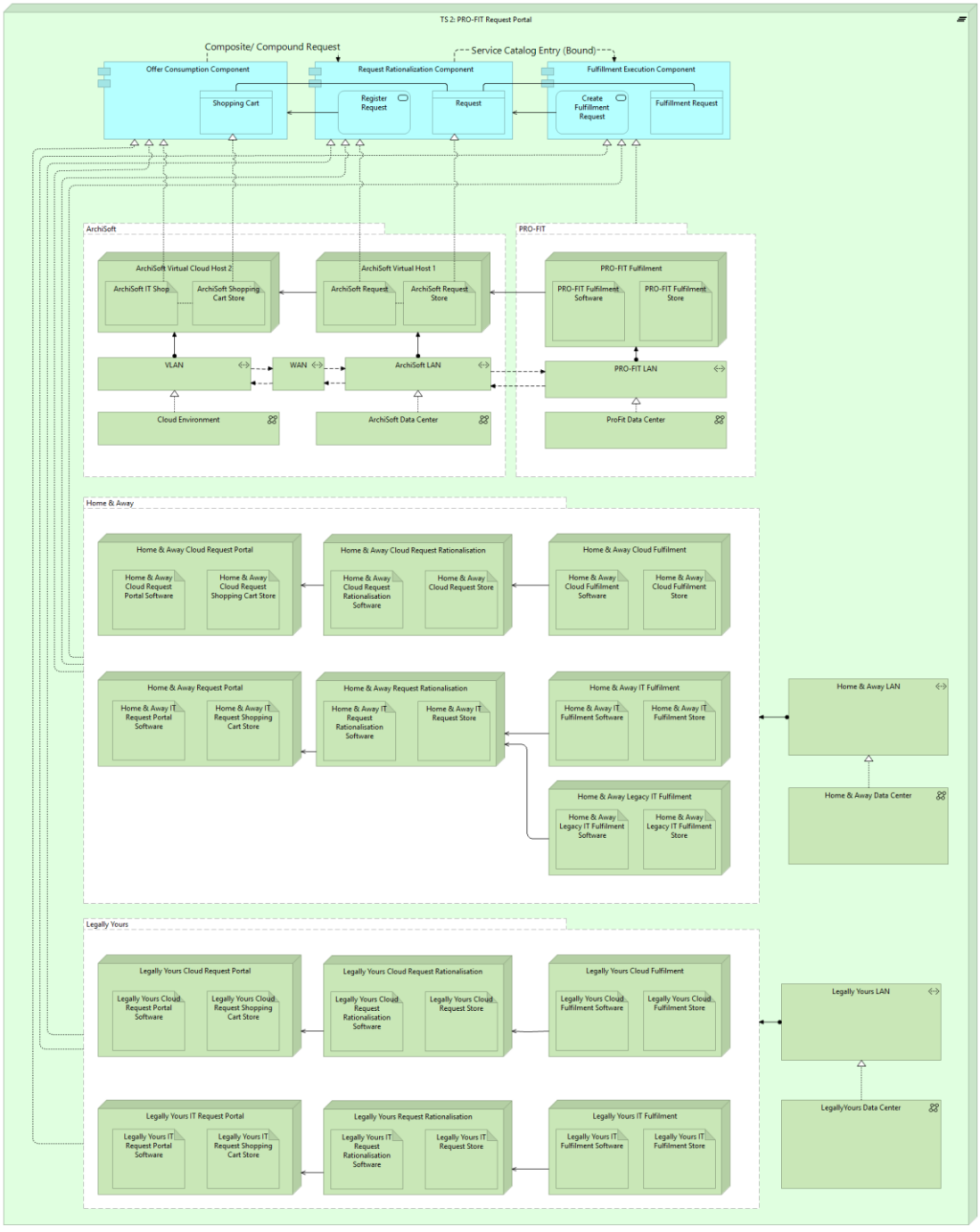


Figure 6: Transition State 2: PRO-FIT IT Service Request Portal

Transition State 3: One Cloud IT Service Request Portal

As soon as the heritage PRO-FIT IT users are migrated to the new IT Service Request Portal (the ArchiSoft IT Shop), the next step is to migrate all the heritage Home & Away and Legally Yours IT users to the new portal, enabling them to use the ArchiSoft IT Shop for requesting cloud services. Requesting Internal IT and the Legacy IT services remains with the heritage Home & Away IT Service Request Portal and the heritage Legally Yours IT Service Request Portal.

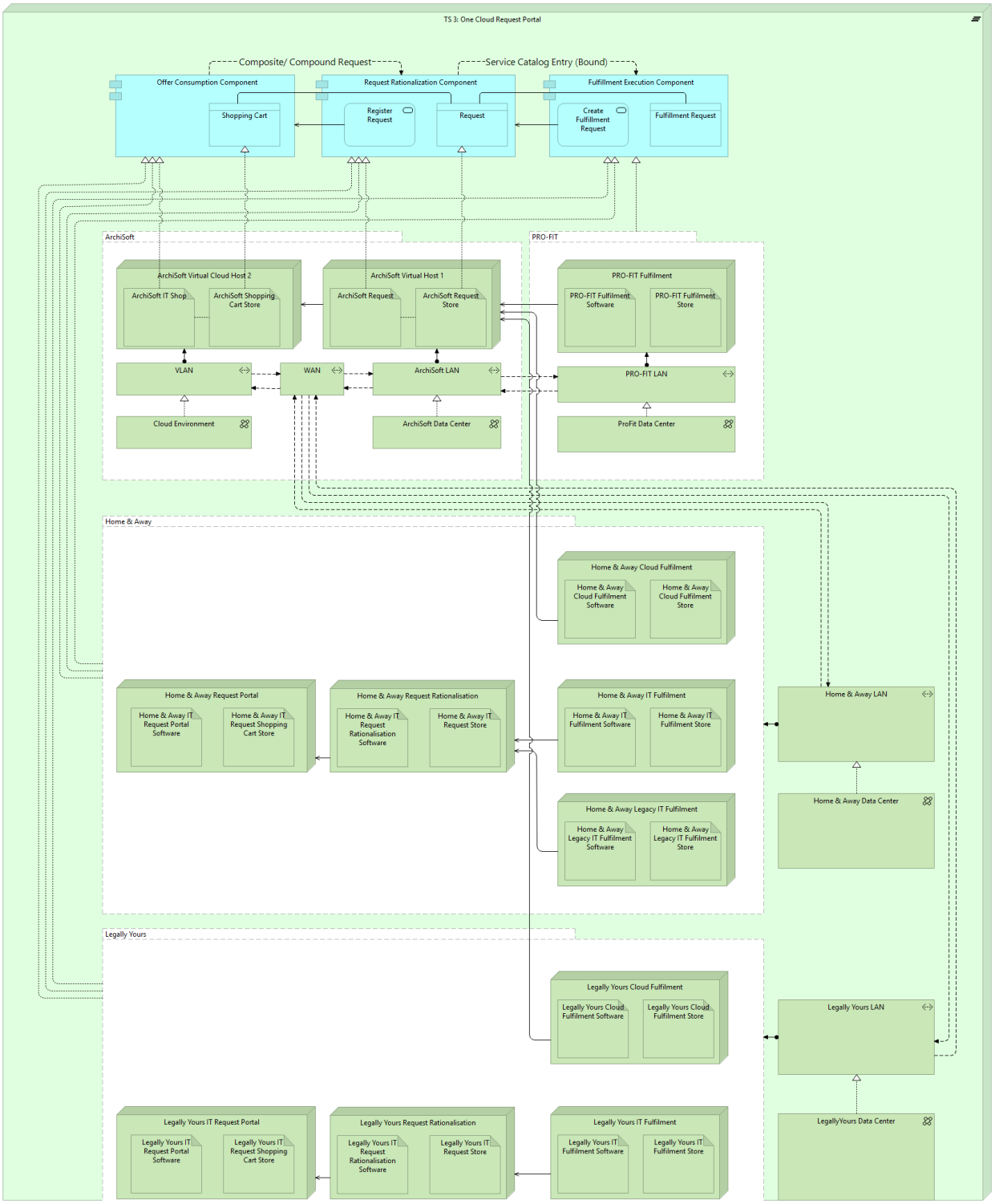


Figure 7: Transition State 3: One Cloud IT Service Request Portal

Target State – Step 4: One Cloud-Based IT Service Request Portal

The last step is to migrate the heritage IT Service Request Portals of Home & Away and Legally Yours to the new cloud-based ArchiSoft IT Shop, replacing and decommissioning all heritage IT Service Request Portals and enabling the new ArchiSoft IT Shop. All fulfillment execution systems are kept in place.

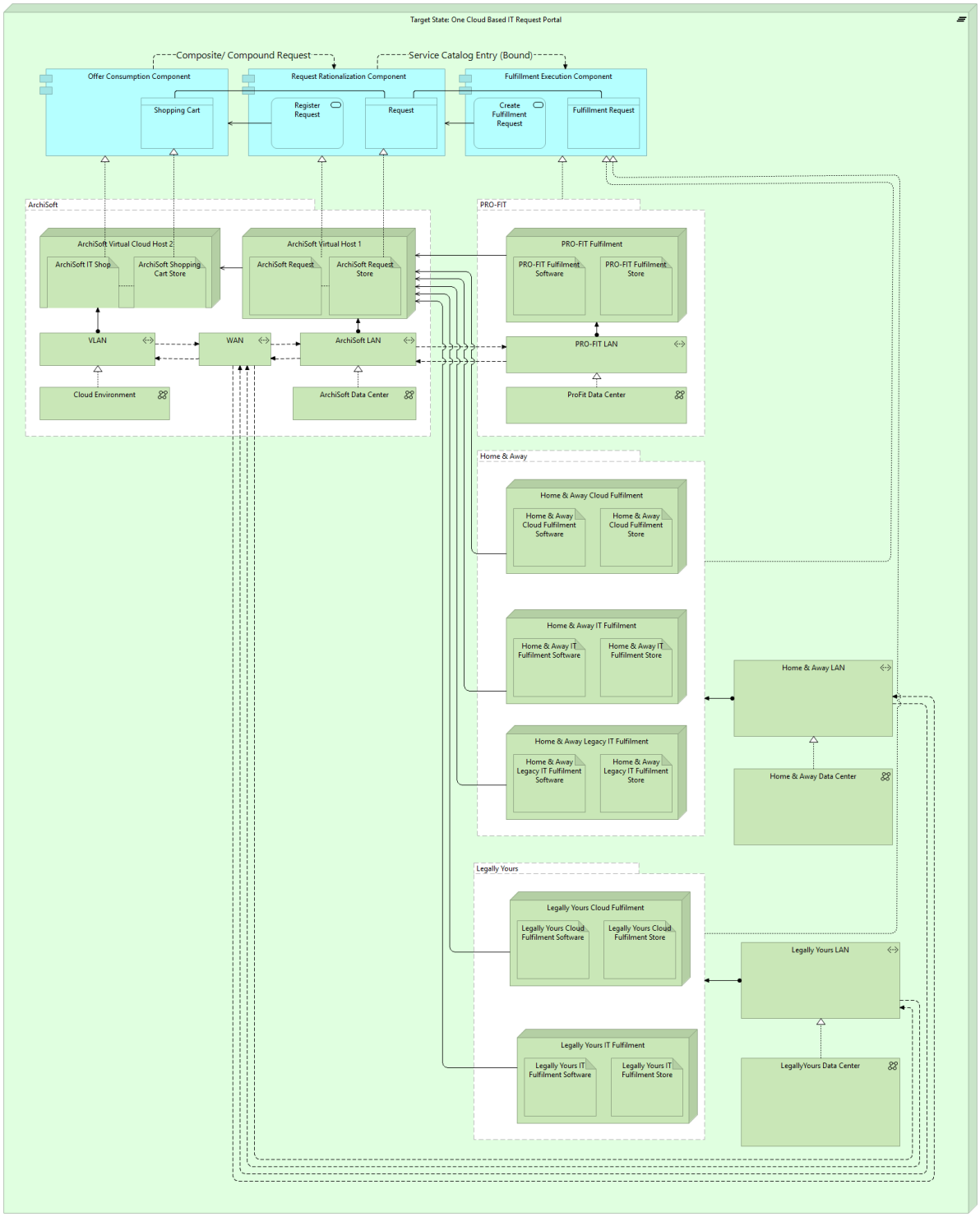


Figure 8: Target State: One Cloud-Based IT Service Request Portal

Modeling an IT Management Landscape

In the previous section, the transition from baseline to target state described a subset of items. To show the possibilities of using the IT4IT Reference Architecture in combination with the ArchiMate modeling language, Figure 9 provides a layered model of the target state.

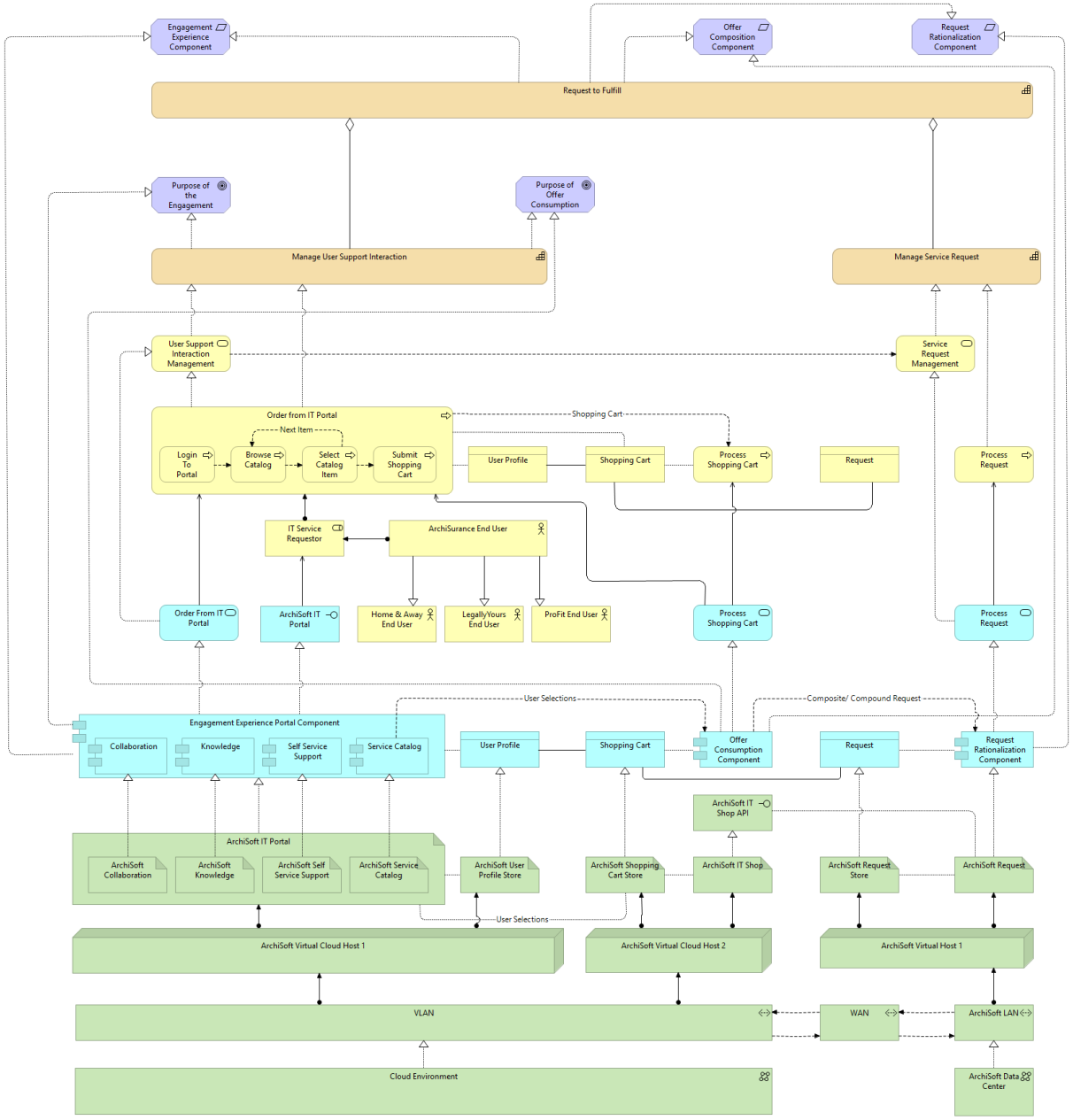


Figure 9: Target State Layered ArchiMate Model

Figure 9 is not intending to be a complete representation, but provides an overview how the IT4IT Reference Architecture can be used to provide guidance and how this can be represented in the ArchiMate language. In Appendix 1, the considerations used to model this diagram are provided.

As can be seen, the IT4IT model can be expanded across the ArchiMate layers, using the IT4IT Reference Architecture as the guidance on the normative elements (for example, application component and data object) as well as non-normative elements (for example, capability).

Conclusion

The ArchiMate language can be used to model IT4IT transformation scenarios.

The ArchiMate language can be used to model the normative (functional components, etc.), non-normative (capabilities, etc.), vendor-specific (system software), transformation (plateaus and motivation for change), and business connection (requirements).

APPENDIX 1: Mapping the IT4IT Reference Architecture to the ArchiMate Language

The IT4IT Reference Architecture itself is described using a bespoke language and identifies five levels (Level 1 to Level 5). For Level 1, 2, and 3, the IT4IT Reference Architecture provides a class model introducing the classes which are used to define the architecture at each of these levels (see Chapter 4 of the IT4IT Reference Architecture).

In the diagrams and tables below, these Level 1, 2, and 3 class models are mapped to a set of corresponding ArchiMate objects.[3] [4]

IT4IT Level 1 Class Model



Page 26 of the IT4IT RA v2.1 Representation within the Archimate Model

Figure 10: Mapping the IT4IT Level 1 Class Model to ArchiMate objects

Table 2: Mapping of the IT4IT Level 1 Class Model to ArchiMate Objects

IT4IT Reference Architecture		**ArchiMate Language**	
Reference Model Concept	**Reference Model Definition**	**Language Concept**	**Language Definition**
Functional Component	A software building block. The smallest unit of technology in the IT4IT Reference Architecture that can stand on its own and be useful as a whole to an IT practitioner (or IT service provider). Functional components must have defined inputs and outputs that are data objects and they must have an impact on a key data object.	Application Component	An encapsulation of application functionality aligned to implementation structure, which is modular and replaceable.

[3] Relationship type classes in the IT4IT class models are represented as a note because a relationship is not an object in the ArchiMate language. The type of relationship to be used between the different objects is provided in the tables and models.

[4] Future versions of the ArchiMate language may contain new ArchiMate language concepts, which could trigger a change in the representation; for example, for IT value chain and IT value stream.

IT4IT Reference Architecture		ArchiMate Language	
Service Backbone Data Object	Key data objects that annotate an aspect of the service model in its conceptual, logical, consumable, or physical state. These data objects and their relationships form the Service Model Backbone which provides a holistic view of a service.	Data Object	Data structured for automated processing.
Lifecycle Data Object	Data or records produced and/or consumed to advance or control the service model as it progresses through its lifecycle phases.	Data Object	Data structured for automated processing.
IT Value Chain	A classification scheme for the set of primary and supporting activities that contribute to the overall lifecycle of creating net value of a product or service offering provided by or through the IT function.	Capability	Represents an ability that an active structure element, such as an organization, person, or system, possesses.
IT Value Stream	Describes the key activities for a discrete area within the IT Value Chain where some unit of net value is created or added to the service as it progresses through its lifecycle.	Capability	Represents an ability that an active structure element, such as an organization, person, or system, possesses.
Relationship	The data objects, combined with their relationships and inter-dependencies, form the “system of record fabric” for IT management (see Figure 21). These relationships are referred to as system of record integrations (record-centric, SoR in short form).	Association	An association relationship models an unspecified relationship, or one that is not represented by another ArchiMate relationship.
N/A	Not available as such in the class model on Level 1. The descriptions of the functional components and data objects provide criteria and data object relationships, which read as requirements.	Requirement	Represents a statement of need that must be met by the architecture.

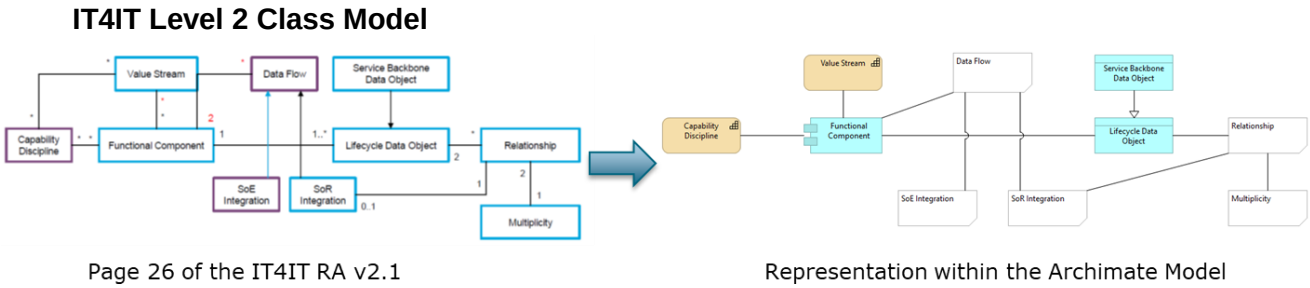


Figure 11: Mapping the IT4IT Level 2 Class Model to ArchiMate Objects

Table 3: Mapping the Additional IT4IT Level 2 Class Model to ArchiMate Objects

IT4IT Reference Architecture		**ArchiMate Language**	
Reference Model Concept	**Reference Model Definition**	**Language Concept**	**Language Definition**
Data Flow	Describes the flow of information between functional components.	Flow	The flow relationship represents transfer from one element to another.
SOR	For a data flow to be refined into a system of record integration, the following conditions must be satisfied: The data flow creates a relationship/dependency between two data objects in the functional components involved in the data flow. The two data objects must have a direct relationship to be maintained. The lifecycles of the two data objects are interlocked. This means that updates to one data object can have consequences on the other data object.	Association (between Artifacts) & Flow (between Application Components)	An association relationship models an unspecified relationship, or one that is not represented by another ArchiMate relationship. The flow relationship represents transfer from one element to another.
SOE	Relationships are also formed between functional components to enable humans and machines to interact with and/or take action on the data objects. These relationships are called system of engagement integrations (experience-centric integration, SoE in short form). These integrations are derived from value stream use-cases and user stories and are offered as guidance rather than prescriptive practices. The intent is to demonstrate to the reader how data object relationships stimulate and/or are affected by human-to-machine and machine-to-machine interactions.	Flow (coloured differently)	The flow relationship represents transfer from one element to another.
Capability Discipline	A capability is the ability of an organization to produce an outcome of value through the utilization of people, process, and technology.	Capability	Represents an ability that an active structure element, such as an organization, person, or system, possesses.
Multiplicity	Multiplicity (sometimes referred to as “cardinality”) describes the number of allowable instances of an element.	N/A	Not defined in the language. Potentially to be added as an attribute of the relation; however, not supported by the ArchiMate language.
N/A	Not available in the Reference Architecture; derived from the original model of Capability Disciplines.	Business Service	Represents an explicitly-defined exposed business behavior. A business service is associated with a value. A business service may serve a business process, business function, or business interaction.

IT4IT Level 3 Class Model

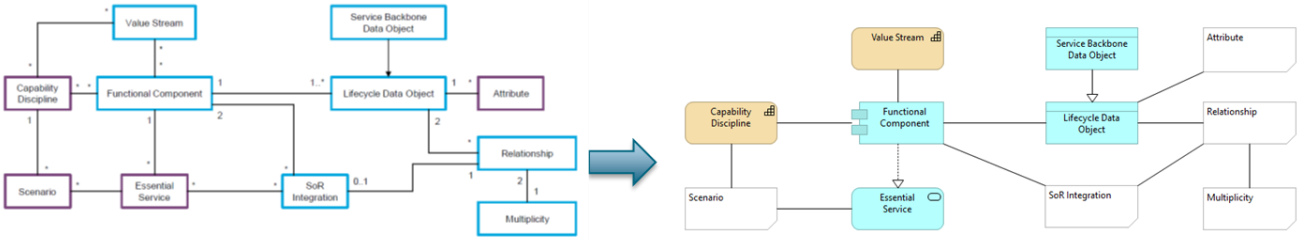


Figure 12: Mapping the IT4IT Level 3 Class Model to ArchiMate Objects

Additional Considerations

Additional ArchiMate Elements Documented in the IT4IT Reference Architecture

The IT4IT Reference Architecture documentation provides a number of descriptions that can be used to define ArchiMate elements. Examples of these descriptions are:[5]

- Each value stream has a number of descriptions currently not separately modeled in the IT4IT Reference Architecture, such as:
  - Objectives [ArchiMate language: Objective]
  - Business Value Proposition [ArchiMate language: Value]
    (Note that the value descriptions in Section 3.3 are not always aligned with the value descriptions in Chapter 5.)
  - Critical Success Factor (CSF) [ArchiMate language: Driver]
  - Key Performance Indicator (KPI) [ArchiMate language: Driver (using specialization: Metric)]
  - Activities (see Section 3.3) [ArchiMate language: Business process using specialization]
- Each functional component has a number of descriptions currently not separately modeled in the IT4IT Reference Architecture, such as:
  - The "Purpose" chapter within the functional component details [Goal]
  - The "Key Data Objects Relationships" and the "Functional Criteria" chapters within the functional component details [Requirement]

Figure 13 gives an example of how such descriptions form the IT4IT Reference Architecture can be modeled in the ArchiMate language.

[5] The ArchiMate language element which could be used is provided between the square brackets [] at the end of the list item.

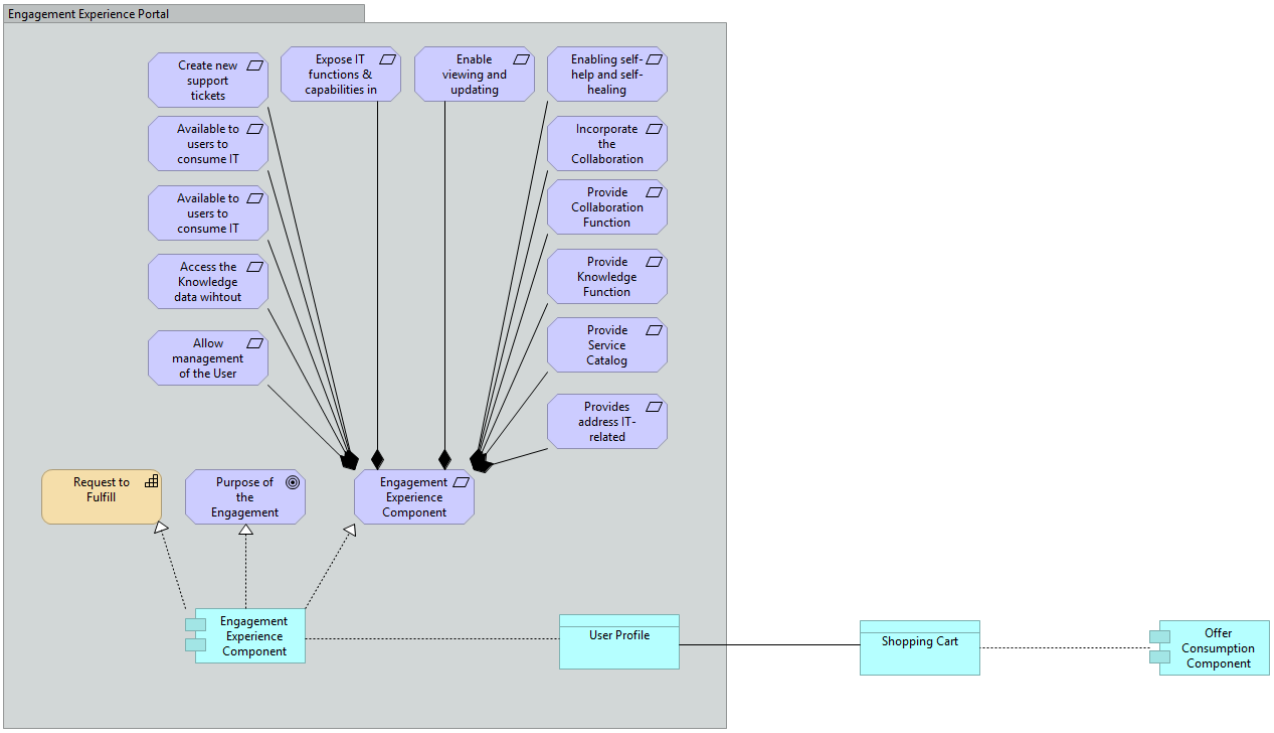


Figure 13: IT4IT Core Elements Represented in the ArchiMate Language

Expanding beyond the IT4IT Reference Architecture Documentation

The ArchiMate language specifies multiple layers (Strategy, Business, Application, Technology, Physical, and Implementation & Migration). The IT4IT Reference Architecture classes are predominantly describing elements within the ArchiMate Application Layer within its set of classes with some additions in the other layers, either as normative or as non-normative classes.

In the following paragraphs, a number of considerations of how to expand beyond the IT4IT Reference Architecture provided classes using the ArchiMate language are provided. In order explain these considerations and to be able to provide an answer to the questions, we will use elements of all ArchiMate layers within the models in this document, complementing the IT4IT model. However, the intent is not to be complete, nor to indicate that alternative choices cannot be used as well. The intent of the document is to validate the possibility to use the IT4IT and ArchiMate standards in the mentioned use-cases.

Expanding into the Strategy Layer

Beyond the definition of the value chain, value stream (both mapped to the ArchiMate Capability element), and the identification of capabilities, the IT4IT Reference Architecture contains no details on the ArchiMate Strategy Layer. This allows usage of the elements of the ArchiMate language without presenting an issue. Figure 14 provides an overview of how to model additional strategy elements.

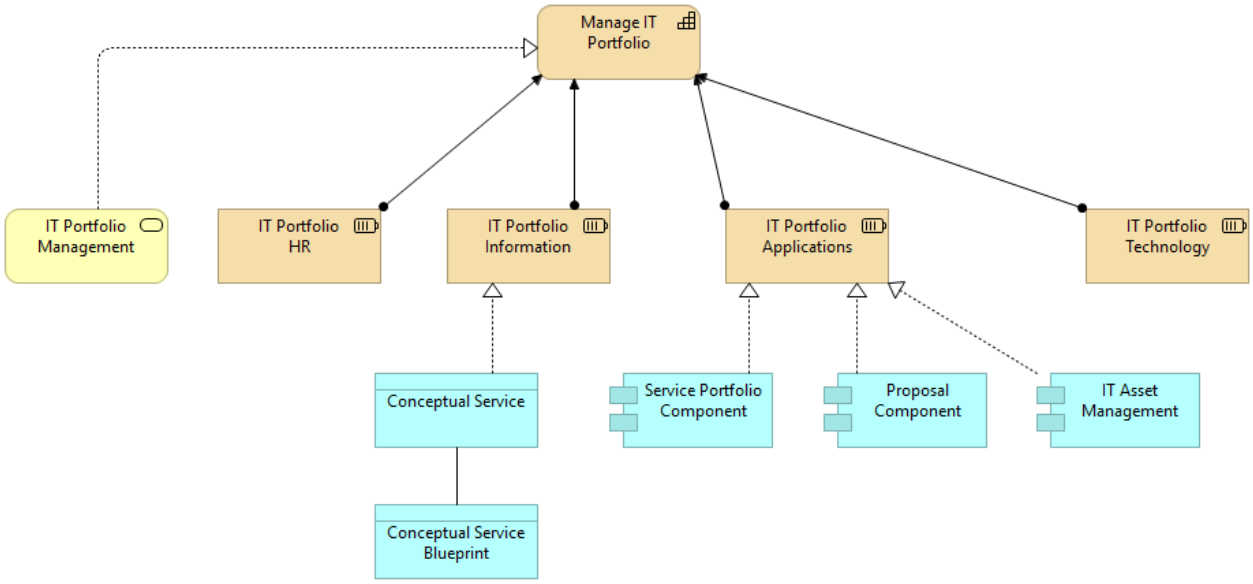


Figure 14: Example Usage of Additional ArchiMate Strategy Layer Elements

Expanding into the Business Layer

Beyond the definition of the value chain, value stream, and the identification of capabilities, the IT4IT Reference Architecture contains no details on the ArchiMate Business Layer. This allows usage of the elements of the ArchiMate language without presenting an issue. As an example, a number of options are provided:

- A Capability Discipline [Capability] being realized by a Process (Step)
- An Essential Service [Application Service] being used in or automating a Process (Step) (servicing a Process)
- System of Insight Integration [Realization between Data Object and Artifact]

This means the ArchiMate elements based on the IT4IT classes can form the basis of a connection to the Business Layer even if they are not defined within the IT4IT Reference Architecture. Typically, such elements are described in scenario documents, and can be found in (standard) repositories or in other documents available. The benefit of this approach is there is no need to re-model the chosen sources provided they are created using the ArchiMate language to be able to re-use the sources in combination with the IT4IT Reference Architecture.

Figure 15 provides an overview of how these elements can be connected to the Core elements.

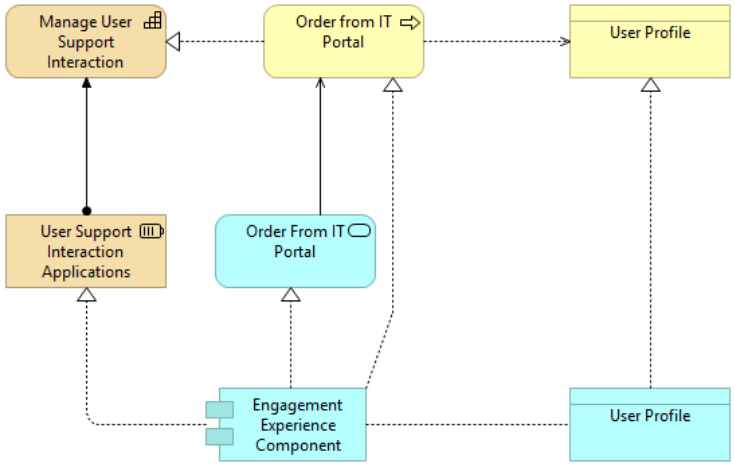


Figure 15: Example Usage of Additional ArchiMate Business Layer Elements

Expanding into the Application Layer

The IT4IT Reference Architecture Classes describe mainly the Application Component [Functional Component], the Data Object [Lifecycle and Service Model data object], and identify Application Services [Essential Service].

The ArchiMate language provides a number of additional elements within the Application Layer and can be added without impacting the intent of the IT4IT Reference Architecture.

Expanding into the Technology Layer

Given the IT4IT Reference Architecture does not prescribe the implementation of the application components (Layer 4 in the IT4IT Reference Architecture is not defined), there is no set of IT4IT Reference Architecture classes which can be mapped to ArchiMate elements. This means there is no specific guidance for usage of the ArchiMate language levels below the Application Level (Technology and Physical). The ArchiMate language provides a number of elements within the Technology Layer and can be added without impacting the intent of the IT4IT Reference Architecture. In fact, the ArchiMate Technology Layer can be used to support the definition of Layer 4 and 5.

For this document, the choices in Table 4 have been made to connect the Technology Layer into the (IT4IT) model:

Table 4: Choices used within this Document to Expand the IT4IT Model into the Technology Layer using ArchiMate Concepts

Language Concept	Language Definition	Implementation Concept	Implementation Concept Definition
Application Component	An encapsulation of application functionality aligned to implementation structure, which is modular and replaceable.	Artifact	Represents a tangible element in the IT world. Artifact is a specialization of technology object. It is typically used to model (software) products such as source files, executables, scripts, database tables, messages, documents, specifications, and model files. An instance (copy) of an artifact can be deployed on a node.
Data Object	Data structured for automated processing.	Artifact	A piece of data that is used or produced in a software development process, or by deployment and operation of an IT system.

Figure 16 shows an example of the link between the IT4IT Reference Architecture elements and the implementations of elements to the ArchiMate language.

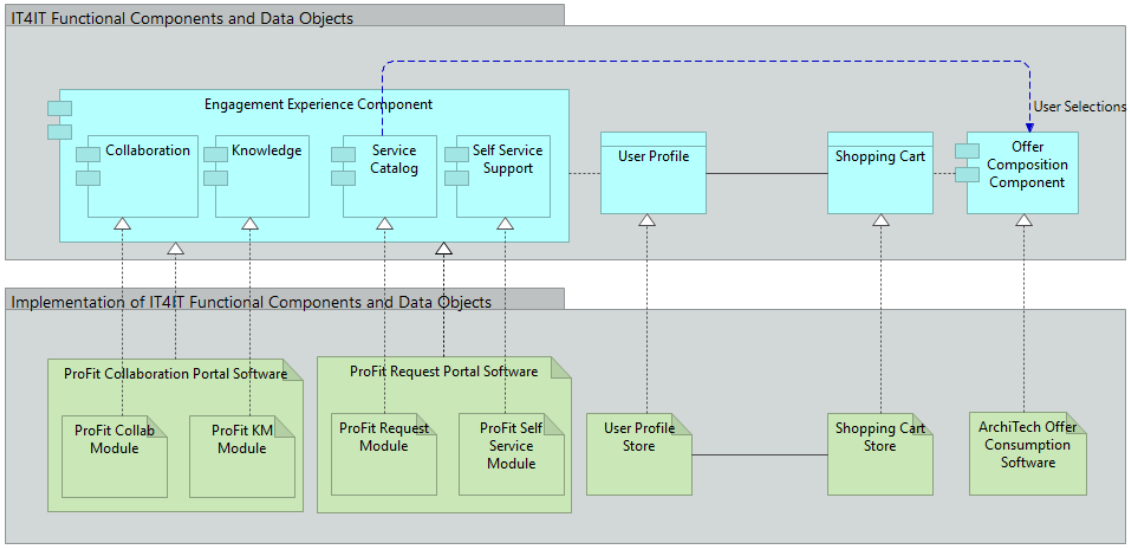


Figure 16: Example Usage of Additional ArchiMate Technology Layer Elements

Expanding into the Physical Layer

Given the IT4IT Reference Architecture does not prescribe the deployment of the application components (Layer 4 and 5 in the IT4IT Reference Architecture are not defined), there is no set of IT4IT Reference Architecture classes which can be mapped to ArchiMate elements in the Physical Layer. This means there is no specific guidance for usage of the ArchiMate language.

The ArchiMate language provides a number of additional elements within the Application Layer and can be added without impacting the intent of the IT4IT Reference Architecture.

Expanding into the Implementation & Migration Layer

Given the IT4IT Reference Architecture does not prescribe classes to support transitioning an architecture, there is no set of IT4IT Reference Architecture classes which can be mapped to ArchiMate elements in the Implementation & Migration Layer. This means there is no specific guidance for usage of the ArchiMate language.

The ArchiMate language provides a number of additional elements within the Implementation & Migration Layer and can be added without impacting the intent of the IT4IT Reference Architecture.

The Solution-Pattern

Figure 17 provides an overview of the solution-pattern used in this document (and as it is used in the ArchiMate model available), providing the overview of the choices made for the modeling.

As can be seen, the model contains both the Normative and the Non-Normative elements as defined in the IT4IT Reference Architecture and is complemented with a number of ArchiMate elements to extend the IT4IT Reference Architecture. Using this meta-model, a model is available as XML using the ArchiMate Open File Exchange and can be found on The Open Group IT4IT Forum website (www.opengroup.org/it4it).

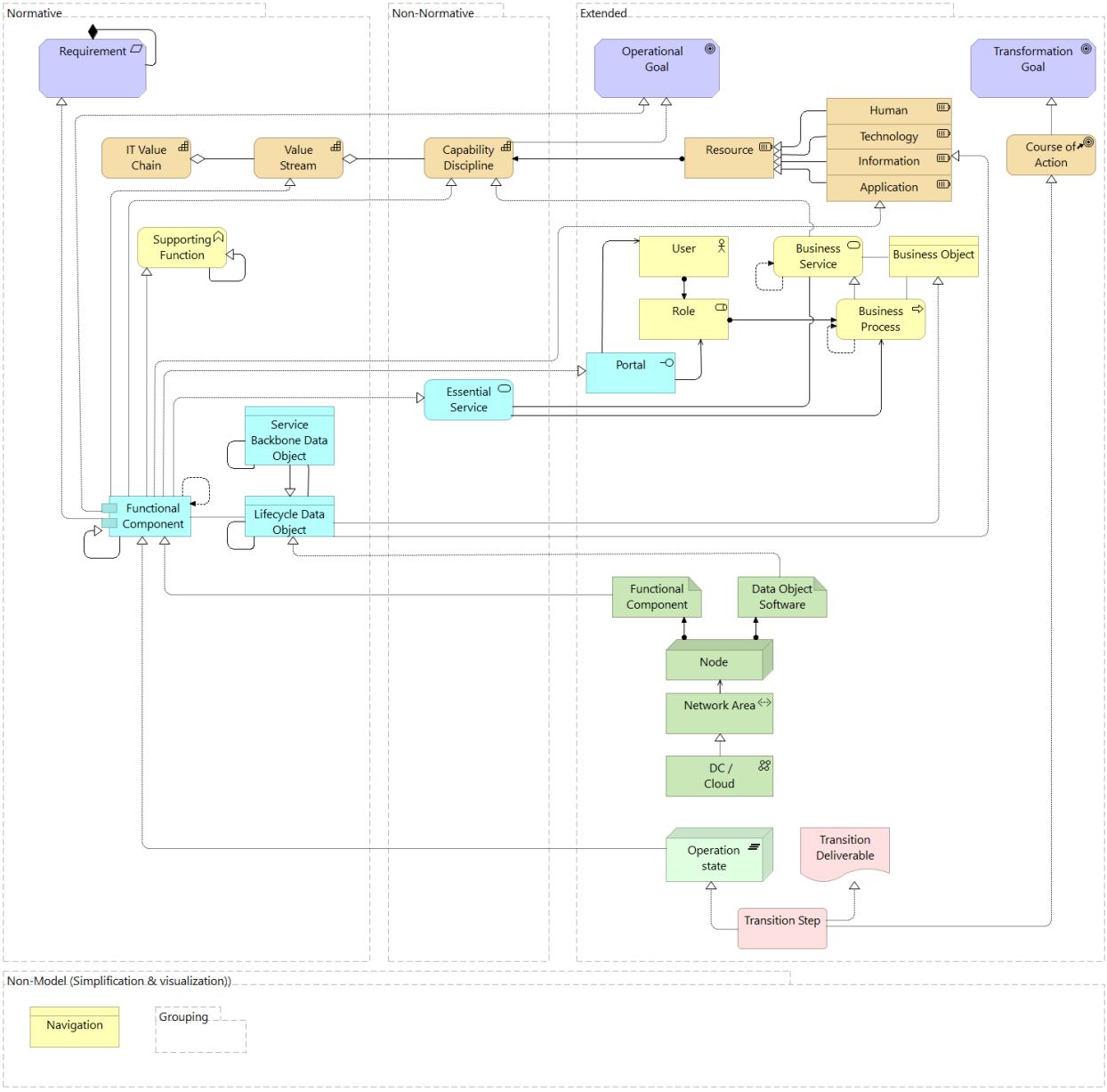


Figure 17: Solution-Pattern

APPENDIX 2: ArchiSurance Health Assessment using the IT4IT Reference Architecture

In order for ArchiSurance to understand the health of their IT4IT Capabilities, ArchiSurance has conducted a health study of its IT4IT components as input to the identification of the needed initiatives.

This is based on the IT4IT Reference Architecture, conducted by the Architecture team of ArchiSurance using interviews and available documentation.

The health assessment has looked at two different aspects:

- Functional components in use
- Systems available

The main focus is on functional components; however, a similar assessment can be done on, for example, Capability, Data, etc. In this case, ArchiSurance decided to focus on the components and systems to minimize the effort but still allow IT Management to make an informed decision.

For the functional components assessment, the following attributes/criteria were used:

- The functional component is in use by all relevant roles
- Usage in process is documented and up-to-date
- Training material for usage is available and up-to-date
- Data managed in the component is accurate and complete

(Score 1-10 per attribute, equally weighted.)

The total score indicates the health of the functional component using colors:

- 1-10 = Red
- 11-20 = Orange
- 21-30 = Yellow
- 31-40 = Green

For the systems assessment, the following attributes/criteria were used:

- Availability and performance of the system
- Under support and/or skilled support team is available
- Number of systems
- Properly integrated with the other components

(Score 1-10 per attribute, equally weighted.)

The total score indicates the health of the functional component using colors:

- 1-10 = Red
- 11-20 = Orange
- 21-30 = Yellow
- 31-40 = Green

This has produced the following results:

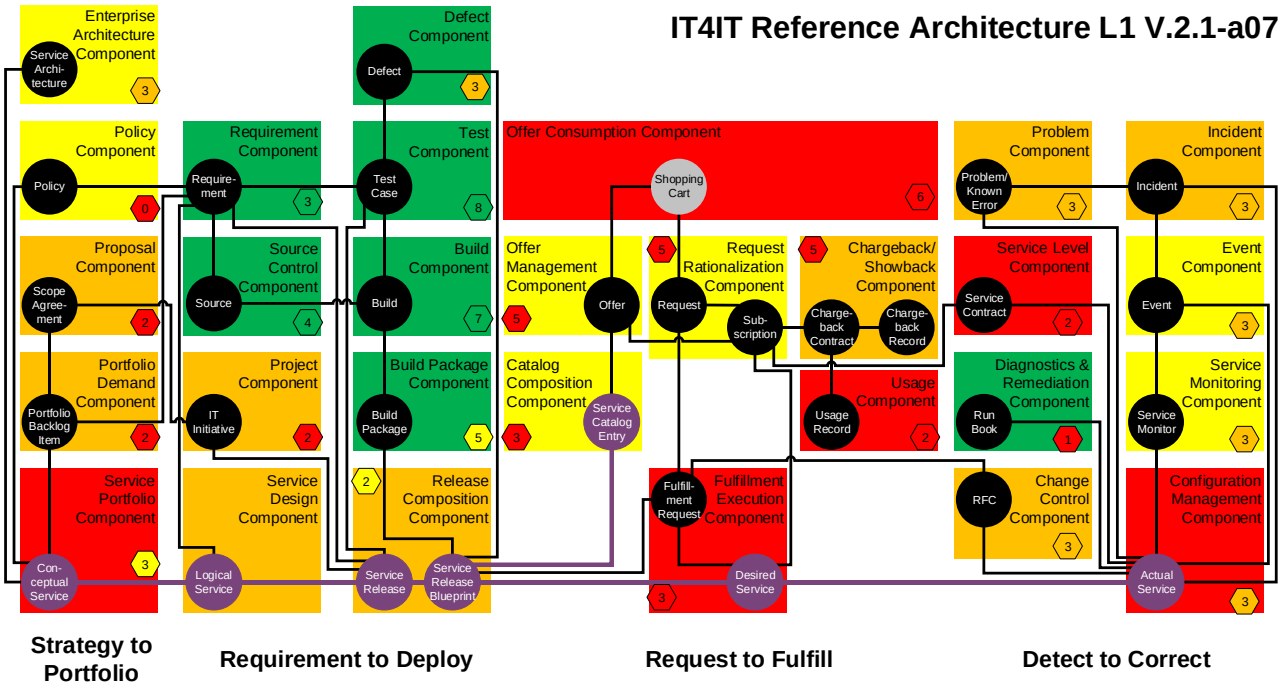


Figure 18: ArchiSurance IT4IT Assessment Results

Figure 18 indicates the combined ArchiSurance results. The coloring is based on the lowest scoring heritage company for an individual functional component. As can be seen, a larger part of the components are supported by more than three systems, indicating the individual heritage companies already had more than one system in use. The assessment shows the biggest issues are in the area of:

- Providing a single IT Service Catalog
- Managing and understanding the configuration of the IT services in production
- Fulfillment of (decomposed elements of) IT requests
- Managing the IT Service Portfolio

The high number of Offer Consumption components (6) indicates a typical situation in organizations: business users do not know where to go when requesting an IT service.

Given the need for consolidation, the individual heritage companies were individually assessed using IT4IT Level 2. The results for the R2F value stream are given below.

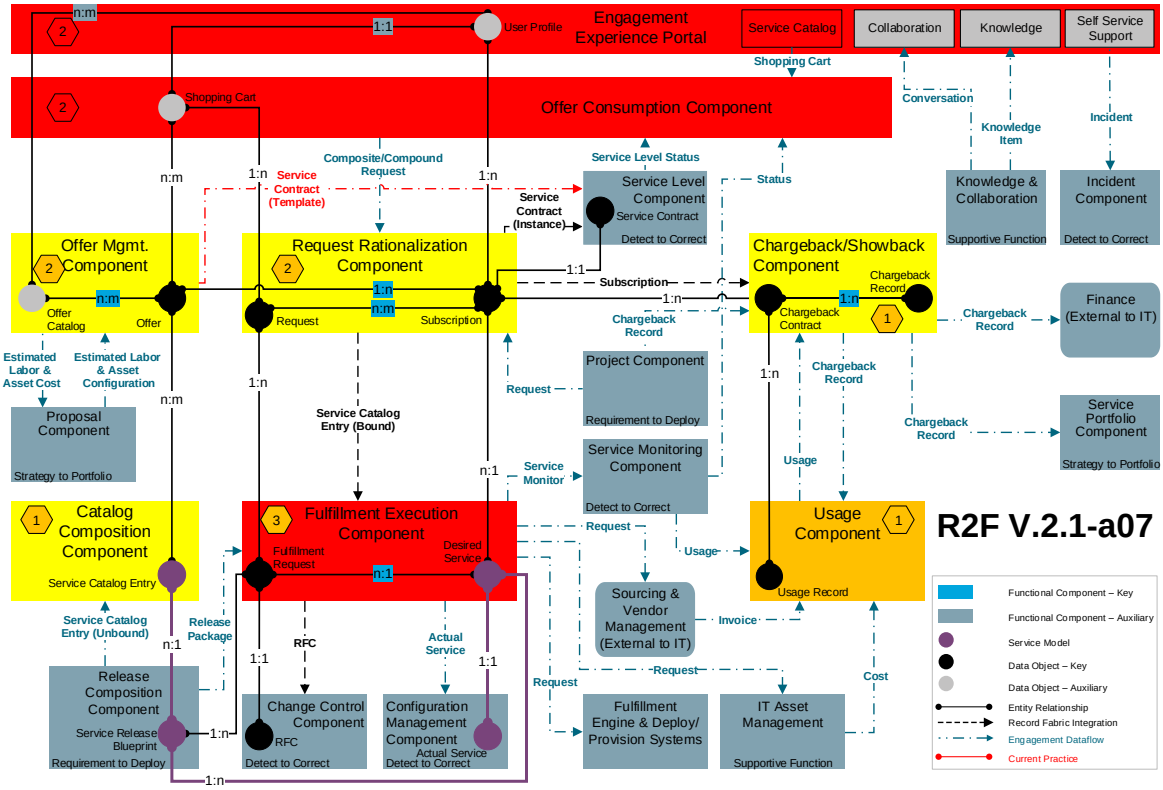


Figure 19: R2F Assessment Home & Away Before Merger

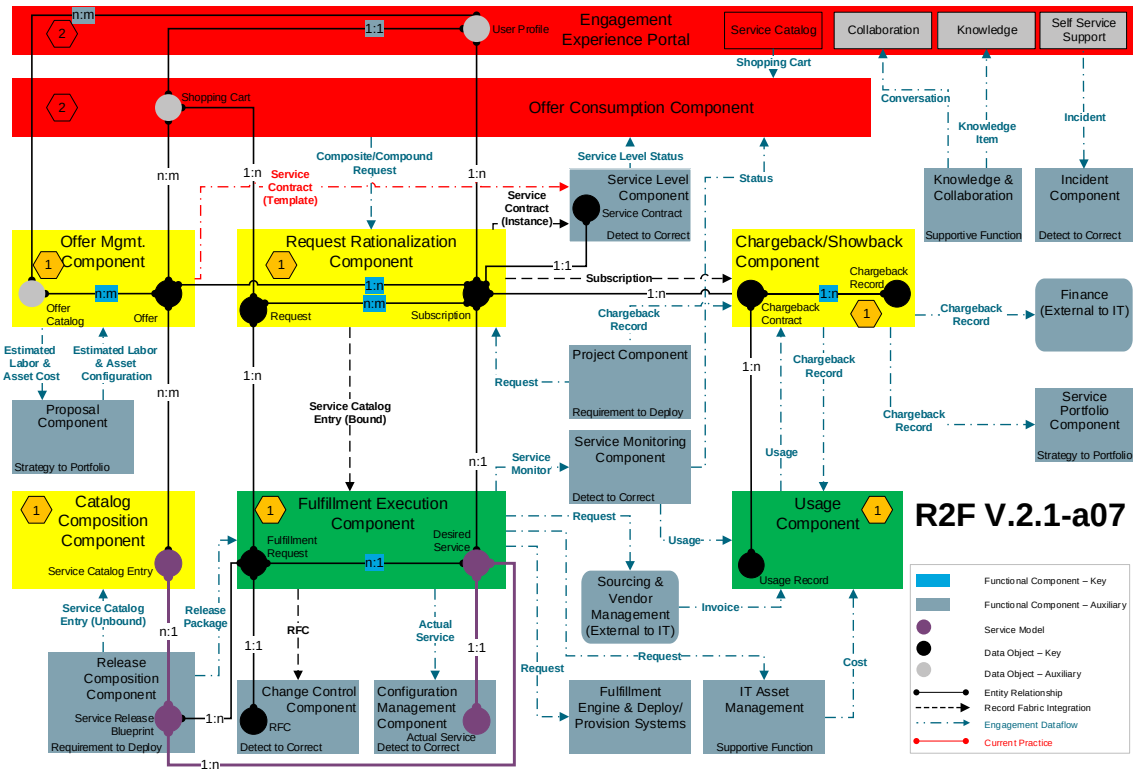


Figure 20: R2F Assessment PRO-FIT Before Merger

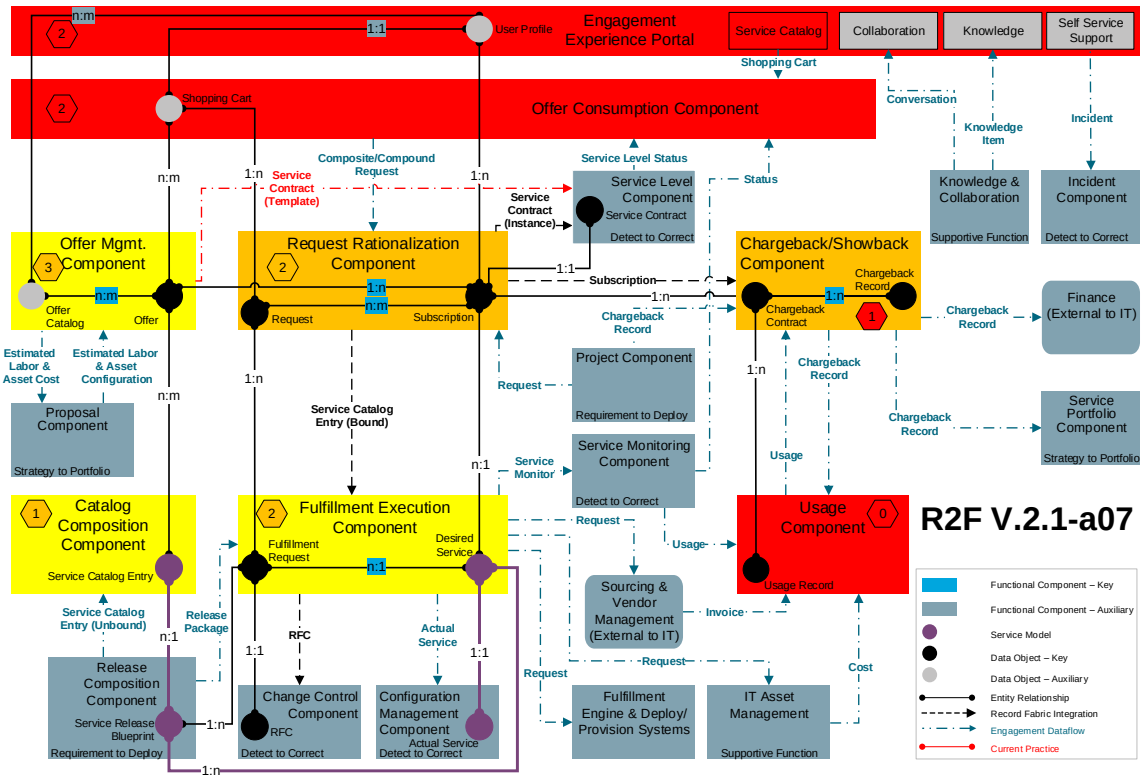


Figure 21: R2F Assessment Legally Yours Before Merger

Out of these assessments, none of the Heritage R2F value streams were implemented without improvement areas, making it difficult for ArchiSurance to propose one of the heritage systems as the target environment, so the proposed approach was to migrate all three heritage systems into a new system, using the latest types of technologies allowing catalog aggregation, minimizing the need for downstream system replacements at the same time.

Given the ArchiSurance "move to the cloud" direction, the architects have proposed to select a system that either runs in the cloud or is provided through the cloud. Given the need for flexibility, the final selection is a container-based IT portal, implemented in a cloud environment.

References

(Please note that the links below are good at the time of writing but cannot be guaranteed for the future.)

- ArchiMate® 3.0.1 Specification, a standard of The Open Group (C179), August 2017, published by The Open Group; refer to: www.opengroup.org/library/c179
- ArchiMate® Model of the IT4IT™ Reference Architecture, Version 2.1; refer to: https://playground.archirepo.com/archirepo/webapp/
- ArchiSurance Case Study, Version 2 (Y163), September 2017, published by The Open Group; refer to: www.opengroup.org/library/y163
- The Open Group IT4IT™ Reference Architecture, Version 2.1, a standard of The Open Group (C171), January 2017, published by The Open Group; refer to: www.opengroup.org/library/c171
- The TOGAF® Standard, Version 9.2, a standard of The Open Group (C182), April 2018, published by The Open Group; refer to: www.opengroup.org/library/c182
- Tool Rationalization using the IT4IT™ Reference Architecture Standard, The Open Group Guide (G191), April 2019, published by The Open Group; refer to: www.opengroup.org/library/g191

About the Author

Kees has worked in the IT Industry for nearly 25 years, as a developer and architect for Mercury Interactive, HP, HPE, and now Fujitsu. Throughout his career, Kees has focused on supporting enterprises in operations and automation of IT service delivery.

Kees was a founding member of the IT4IT Standard development team. From its inception, he has driven the creation of an IT4IT Reference Architecture model using the ArchiMate language.

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- Linda Kavanagh
- Cathy Fox

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